

COURSE 7

SQL relational language

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SQL Features

- is a *relational language* built on *relational expressions transformation*
- is a *descriptive programming language*
- has an *interpretative* working style
- has great *retrieval power*
- is *the standard relational language*
- is part of the core architecture of a relational DBMS, thus including Oracle kernel, which ensures portability for relational database.

SQL language elements

The expressions used in commands are built using the operators and operands. The expression can be of several types, depending on the type *of operators* that appear in the expression:

- *arithmetic* (+, -, *, /),
- *comparison* (=, <>, >, > =, <, <=),
- *logical* (AND, OR, NOT),
- *special* (LIKE, IN, BETWEEN),
- *relational* (projection, selection, join, union, intersection, minus).

SQL language elements

The *operands* of expressions are: constants, (system – indicate how the system is configured, and programmer) variables, (system and programmer) functions, tables (for relational expressions).

Comments can be written in three ways:

- `/ * Text * /` desired text is delimited by the two characters specified;
- `--` at the beginning of the comment. This text cannot extend to a new line.
- `REM[ARK]` command which marks the beginning of a comment; This command cannot be used in PL/SQL blocks while the other two types of comments can.

SQL commands

Controls for setting options

SET	Assigns values / changes status to parameters / variables of the system
ACCEPT	Creates a programmer variable and assigns a value taken from the standard input peripheral - keyboard
PROMPT	Writes on standard data output peripheral - screen
DEFINE	Defines a string programmer variable in internal memory
UNDEFINE	Deletes a previously defined programmer variable
DESCRIBE	Displays a table's data structure from the database dictionary
HELP	Displays information about the specified system command / clause / function

SQL commands

Commands for defining data

CREATE PARTITION	Creates a user partition (to restrict access to the database)
ALTER PARTITION	Updates an existing partition
CREATE TABLESPACE	Defines a model for allocating physical space for a table / index / cluster
ALTER TABLESPACE	Upgrades to a physical space allocation model that already exists
DROP TABLESPACE	Deletes a physical space allocation model that already exists
CREATE DATABASE	Creates files in the physical structure of RDB: data files (for: basic tables, database dictionary, etc.), control files, files for recovery, and opens the database
CREATE TABLE	Creates the data structure for a table - describing attributes and integrity constraints, which are stored in the database dictionary
ALTER TABLE	Updates the data structure for a table
DROP TABLE	Clears the data structure for a table with related integrity constraints and indexes
CREATE INDEX	Creates an index file attached to the specified table
VALIDATE INDEX	Checks the integrity of a file index
DROP INDEX	Deletes the specified index
CREATE OR REPLACE VIEW	Creates a virtual table / view in a retrieval request from one or many tables
DROP VIEW	Deletes a virtual table
CREATE SYNONYM	Creates a synonym / alias for a table or view
DROP SYNONYM	Removes a synonym
CREATE CLUSTER	Creates a cluster with data from one or many tables
ALTER CLUSTER	Upgrades a cluster
DROP CLUSTER	Deletes a cluster

SQL commands

Commands for updating

INSERT	Adds, at the end of a table, a tuple (using the VALUES clause) or a set of tuples (using SELECT)
UPDATE	Modifies tuples from a specified table
DELETE	Deletes tuples from a specified table
SET TRANSACTION	Sets the working conditions for a transaction
SET AUTOCOMMIT	Sets automatic completion (ON) or not (OFF) as default transaction
COMMIT	Completes all previous transactions
ROLLBACK	Cancels all previous transactions that were not completed
SAVEPOINT	Marks save-points in a transaction (sub-transaction)

SQL commands

Commands for data querying

COPY	Copies data from one source to one destination table
SHOW	Displays the values of system variables
COMPUTE	Performs calculations as indicated in arithmetic expression
SELECT	Retrieves simple or complex queries, using the appropriate clauses. In Oracle implements the six basic relational operators: projection, selection, join, union, minus, intersection.

```
BREAK ON JOB_ID;  
COMPUTE SUM LABEL 'TOTAL' OF SALARY ON JOB_ID;  
SELECT JOB_ID, LAST_NAME, SALARY  
FROM EMPLOYEES  
WHERE JOB_ID IN ('AC_MGR', 'SA_MAN')  
ORDER BY JOB_ID, SALARY;
```

Some system functions

- *Aggregation functions:* ABS () - absolute value, MOD (a,b) – returns the remainder of a divided by b , ROUND () - rounding a value to a desired number of decimal places, TRUNC () - truncate a value to a desired number of decimal places, SQRT () - square root.
- *Character functions:* ASCII () - ASCII value of a specified character, CHR () - character corresponding to ASCII value, LOWER () - converts a string to lowercase, UPPER () - converts a string to uppercase, LENGTH () - provides the length of the string, SUBSTR () - extract a sub-string from a given string.
- *Group functions:* AVG () - the arithmetic mean of a number of elements, SUM () - sum of values, COUNT () - counts the elements that satisfy a certain condition, MAX () - maximum value of a range of elements, MIN () - the minimum value of a number of elements.
- *Conversion functions:* TO_CHAR - converts a number or a date value to a string value, TO_DATE - converts a string value to DATE data type value. There are a number of functions for processing the dates.

DB-Engines Ranking

423 systems in ranking, November 2024

Rank			DBMS	Database Model	Score		
Nov 2024	Oct 2024	Nov 2023			Nov 2024	Oct 2024	Nov 2023
1.	1.	1.	Oracle 	Relational, Multi-model 	1317.01	+7.57	+39.98
2.	2.	2.	MySQL 	Relational, Multi-model 	1017.80	-4.95	-97.44
3.	3.	3.	Microsoft SQL Server	Relational, Multi-model 	799.81	-2.28	-111.61
4.	4.	4.	PostgreSQL 	Relational, Multi-model 	654.34	+2.18	+17.48
5.	5.	5.	MongoDB 	Document, Multi-model 	400.93	-4.28	-27.62
6.	6.	6.	Redis 	Key-value, Multi-model 	148.64	-0.99	-11.38
7.	7.	 11.	Snowflake 	Relational	142.50	+1.90	+21.50
8.	8.	 7.	Elasticsearch	Multi-model 	131.64	-0.20	-7.98
9.	9.	 8.	IBM Db2	Relational, Multi-model 	121.74	-1.03	-14.26
10.	10.	 9.	SQLite	Relational	99.49	-2.43	-25.09

416 systems in ranking, November 2023

Rank			DBMS	Database Model	Score		
Nov 2023	Oct 2023	Nov 2022			Nov 2023	Oct 2023	Nov 2022
1.	1.	1.	Oracle 	Relational, Multi-model 	1277.03	+15.61	+35.34
2.	2.	2.	MySQL 	Relational, Multi-model 	1115.24	-18.07	-90.30
3.	3.	3.	Microsoft SQL Server 	Relational, Multi-model 	911.42	+14.54	-1.09
4.	4.	4.	PostgreSQL 	Relational, Multi-model 	636.86	-1.96	+13.70
5.	5.	5.	MongoDB 	Document, Multi-model 	428.55	-2.87	-49.35
6.	6.	6.	Redis 	Key-value, Multi-model 	160.02	-2.95	-22.03
7.	7.	7.	Elasticsearch	Search engine, Multi-model 	139.62	+2.48	-10.70
8.	8.	8.	IBM Db2	Relational, Multi-model 	136.00	+1.13	-13.56
9.	9.	 10.	SQLite 	Relational	124.58	-0.56	-10.05
10.	10.	 9.	Microsoft Access	Relational	124.49	+0.18	-10.53

<https://db-engines.com/en/ranking>

<https://trends.google.com/trends/explore?date=all&geo=RO&q=oracle,mysql,mongo%20db>